Python source

```
import arrus.numpy as anp

with arrus.Graph() as g:
  f   = anp.tx_rx_seq(@indices)
  b   = anp.beamform(f)
  img = anp.scan_convert(b)

session.compile(g)
session.upload()
session.start()
```

[compile]

IR stack

```
Level 1 --- arrus dialect
f   = arrus.tx_rx_seq(@indices)
b   = arrus.beamform(f)
img = arrus.scan_convert(b)
```

```
Level 2 --- us4r + gpu
w = us4r.rebind_wiring(@indices)
f = us4r.fire_sequence(w)
b = gpu.beamform(f)
```

```
+ graph pruning / DCE /
canonicalization
```




```
Device asm / kernels
us4r: TX[i]{ap, dl, pl}
DMA descriptors [0..K]
GPU PTX / SASS
```

[upload]

Device memory

```
Code memory
TX/RX linked list [0..191]
GPU kernels loaded
DMA descriptor table
```

```
Variable memory
@indices = [3, 5, 8, 13, ...]
@sos     = 1540.0
@tgc     = [...]
```

[start]

Running system

